

The weighted analytic coefficient-pair space

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Abstract

Astra #11 rank 2. The Taylor data (x, y) of phase and amplitude are realised as a single point of Mathlib's Banach space $\ell^1((\mathbb{N}^2) \sqcup (\mathbb{N}^2), \mathbb{R})$ after the rescaling $a(\text{inl } k) = x_k \rho^{|k|}$, $a(\text{inr } k) = y_k \rho^{|k|}$ (`CoeffPair`, `toX`, `toY`, `ofPair`, with round trips and $\|a\| = \|x\|_\rho + \|y\|_\rho$). The norm dominates both weighted norms, so by unit 145 the canonical coefficient maps `coeffA`, `coeffB`: `CoeffPair` $\rightarrow \mathbb{R}$ are continuous (locally Lipschitz) and Borel measurable. This is the domain on which convergence in distribution of the coefficient data will be stated. Zero sorry/axiom.

1 The things formalised

```
abbrev PairIdx := ( × ) ( × )
abbrev CoeffPair := lp (fun _ : PairIdx => ) 1 -- + Borel instance
def toX ( ) (a : CoeffPair) : × → := fun k => a (Sum.inl k) / ^ (k.1 + k.2)
def toY ... (Sum.inr)
theorem norm_eq_tsum_abs (a) : ||a|| = ' i, |a i|
theorem wsummable_toX / wsummable_toY ( ) (h : 0 < ) (a) : WSummable (toX a)
theorem wnorm_toX_le_norm / wnorm_toY_le_norm : wnorm (toX a) ||a||
theorem wnorm_toX_sub_le_dist : wnorm (fun k => toX a k - toX b k) dist a b
def ofPair ( ) (h) (x y) (hx hy) : CoeffPair -- Sum.elim of the rescaled arrays
theorem toX_ofPair / toY_ofPair / ofPair_toXY -- round trips
theorem norm_ofPair : ||ofPair h x y hx hy|| = wnorm x + wnorm y
def coeffA ( ) (h h k k) ( ) (a) :=
  canonA ... (fun i j s => ampCoeff (toX a) (toY a) (i,j) s)
def coeffB ( b ) ... := canonB ... (anaFaceU ...) (anaFaceV ...) ...
theorem continuous_coeffA ( r) (h h k k) (h h hr0 hr hk hk) ( ) (h) :
  Continuous (coeffA h h k k )
theorem continuous_coeffB ... (hb hb hbr hr ...) : Continuous (coeffB b h h k k )
theorem measurable_coeffA / measurable_coeffB
```

2 Mathematics

The weighted norm $\|x\|_\rho = \sum |x_k| \rho^{|k|}$ is the ℓ^1 norm of the rescaled array, so the space of weighted-summable arrays is isometric to $\ell^1(\mathbb{N}^2)$; a pair of arrays is a point of ℓ^1 over the disjoint union of two copies of $\mathbb{N}^2$, with norm the sum of the two weighted norms. Working in a single ℓ^1 space (rather than a product of two) sidesteps the product-Borel- σ -algebra question: the Borel σ -algebra of a product of metric spaces equals the product σ -algebra only under second countability, which Mathlib does not provide for `lp` in general. Continuity of A_α, B_α follows from the unit-145 Lipschitz bound on the unit ball around each point (with $M = \|a_0\| + 1$ as the common bound) via `continuousAt_of_locally_lipschitz`; measurability is then automatic.

3 Paper-facing meaning

Astra #11 recommended defining the coefficient topology through the arrays and treating the bridge to the paper's C^ω topology as an unformalised compatibility statement. The present space is that definition: a separable Banach space of Taylor data in which the $d = 2$ canonical coefficients are continuous functions, ready for the continuous mapping theorem (`TendstoInDistribution.continuous_comp`) in the next unit.

4 Lean notes

- `Mem p` for $p = 1$: `(mem p_gen_iff (p := 1) (by simp)).1 (lp.mem p a)` then `simpa` turns $\|a_i\|^1$ into $|a_i|$; `lp.norm_eq_tsum_rpow (by simp)` plus `simp` gives $\|a\| = \sum |a_i|$.
- Summability over a sum type: `Summable.sum f h h` (from `Multipliable.sum` by `to_additive`) and `Summable.tsum_sum`; there is no `Summable.sum_elim`. Row inequalities against the total use `Summable.tsum_le_tsum_of_inj` with `Sum.inl_injective`.
- The coercion of `f, hf : lp E p` to a function is `Subtype.val`, so `change` exposes `f k` directly; membership `f : lp E p` is `Mem p f p` by `change`.
- A `Borel` instance on the abbreviation `CoeffPair` is `noncomputable instance : MeasurableSpace CoeffPair := borel CoeffPair with BorelSpace := rfl`.